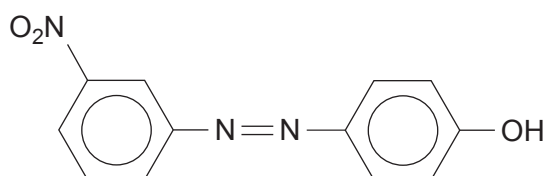


5. The formula of an azo dye, produced by reacting a diazonium compound with phenol, is shown below.



Write the formula of the amine that is used to produce this diazonium compound.

[1]

6. Ethanal reacts with hydrogen cyanide to give 2-hydroxypropanenitrile, $\text{CH}_3\text{CH}(\text{OH})(\text{CN})$.

(a) State the type of mechanism occurring during this reaction.

[1]

.....

(b) Draw the structures of the two enantiomers of 2-hydroxypropanenitrile.

[1]

